Shadows of the Fourth Dimension

Deltahedra that join differently: the triangle-faced shadows and slices of the 5-, 16-, 24- and 600-cell



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The ideas are DICTO's. Computation and proofs with Claude (Anthropic).

Abstract. A deltahedron is a solid whose faces are all equilateral triangles; only eight of them are convex. Four regular solids of four-dimensional space are also built from triangles: the 5-cell, the 16-cell, the 24-cell and the 600-cell. DICTO's idea was that their three-dimensional shadows and slices might stand closer to a new family of deltahedra — same pattern of triangles, different shape — than to the eight regular ones, and that such variants would join to other pieces differently. We compute every shadow and slice of the four solids along their four main directions exactly, in numbers of the form a + b√5, and find deltahedra of two new kinds. (a) Two shadows are variants of convex deltahedra, and each joins differently from its regular counterpart: the 5-cell's edge-first shadow (TBP-5) has the pattern of the triangular bipyramid J12, and three copies of it close round an edge; the 16-cell's edge-first shadow (OCT-16) has the pattern of the octahedron, and it fills space on its own. (b) Four shadows and slices of the 600-cell can be rebuilt exactly with every edge equal, keeping their pattern of triangles, and all four are non-convex: a 12-dented icosahedron; the excavated and the augmented dodecahedron; Shadow-17, two diminished icosahedra (J11) glued on their pentagons, identified through DICTO's idea of cutting the pattern in halves; and Slice-48, a solid of 92 triangles with tetrahedral symmetry, found by solving the 600-cell's six large slices exactly with their full symmetry. Supporting results: apart from copies of the solids' own cells and vertex figures, no shadow, layer or slice is a regular deltahedron (Proposition 1), and the other five large slices have no equilateral form with their full symmetry (§6).

1. The idea

DICTO's idea (27 October 2026) was that "4th dimensional shadows may more closely represent any new deltahedron type family than they do the original ones", and that such variants, set against the regular ones, would be "different shapes with different attachability". On the same day a search for new deltahedra glued from DICTO's pieces (AXE, CLEO, Cleo-Icos, Admiral Cap, Tricap and others; up to 4 pieces of all kinds, up to 5 of DICTO's) had found 772 solids with all faces equilateral, every one of them a gluing of regular tetrahedra, octahedra and icosahedra: no new deltahedron and no variant. The fourth dimension was the next place to look, and it gave the following.

Main result. The triangle-faced shadows and slices of the 5-, 16-, 24- and 600-cell give deltahedra of two new kinds.
(a) **Variants that join differently** (§4). The 5-cell's edge-first shadow (TBP-5) has the pattern of the triangular bipyramid J12 with unequal edges, and three copies of it close face to face round an edge. The 16-cell's edge-first shadow (OCT-16) has the pattern of the octahedron with unequal edges, and it fills space on its own.

(b) **Equal-edged solids from the 600-cell** (§5, §6). Four of the 600-cell's shadows and slices can be rebuilt exactly with every edge the same length, keeping their pattern of triangles: the vertex-first shadow (42 vertices, 80 triangles) as a 12-dented icosahedron; the slices at the dodecahedron height (32 vertices, 60 triangles) as the excavated and the augmented dodecahedron; the outer edge-first slices (17 vertices, 30 triangles) as Shadow-17, two J11 glued on their pentagons; and the 48-vertex cell-first slice (92 triangles) as Slice-48, with tetrahedral symmetry. All four forms are non-convex.

2. Method

All arithmetic is exact in $\mathbb{Q}(\sqrt{5})$ (numbers $a + b\sqrt{5}$, a and b rational, with exact signs); floating point only proposes convex hulls, and every hull face is then checked exactly (all points on one side, every edge in two faces, Euler characteristic 2). For each solid and each of four directions — through a vertex, the middle of an edge, the centre of a triangle and the centre of a cell — we take the **shadow** (the outline of the projection into the 3-space at right angles), the **layers** (the outline of the vertices at one height) and the **slices** (the true cut by a 3-space at right angles, through the centre and at every vertex height: the vertices on it plus every edge crossing it). This gives 223 records and 157 distinct solids, 27 of which have all faces triangles. Each of the 27 is compared with the eight convex deltahedra by graph isomorphism; for the patterns that match none of the eight, an equilateral form is sought (least squares on the edge lengths, as in Tsuruta et al. 2014) and confirmed exactly where found.

The shadows and slices themselves are convex, as every projection and cut of a convex solid is, and in general their edges are unequal. The question for each is whether its *pattern* of triangles can be rebuilt with all edges equal. Where that needs some edges folded inwards, the equilateral form is non-convex: the dents belong to that form, not to the 4D solid.

3. Results

The table lists the all-triangle shadows and slices other than regular tetrahedra, octahedra and icosahedra, grouped by pattern, with the equilateral form found for each pattern.

4D source	Solid	Pattern	Edge ratio	Equilateral form
5-cell	edge-first shadow (TBP-5)	triangular bipyramid J12	$\sqrt{(12/7)}$ = 1.309	variant of J12
16-cell	edge-first shadow = central slice (OCT-16)	octahedron	$2/\sqrt{3}$ = 1.155	variant of the octahedron
600-cell	vertex-first shadow = central slice (42, 80)	pentakis icosidodecahedron	$2/\sqrt{3}$	12-dented icosahedron (exact)
600-cell	vertex-first slices at the dodecahedron height (32, 60)	pentakis dodecahedron	1.144	augmented dodecahedron; excavated dodecahedron (both exact)
600-cell	edge-first slices (17, 30): Shadow-17	none of the 8	2.927	two J11 on their pentagons (exact)
16-, 600-cell	face- and edge-first (8 or 12 vertices)	hexagonal, decagonal bipyramids	1.41–2.29	none can exist (Prop. 2)
600-cell	six large slices (28–72 vertices)	none of the 8	2.29–4.81	§6

All other all-triangle results are the regular tetrahedron, octahedron or icosahedron, and each is a cell or vertex figure of its 4D solid. Two supporting propositions fix what the table leaves out.

> **Proposition 1.** No shadow, layer or slice of the 5-, 16-, 24- or 600-cell along the four main directions is a regular deltahedron, except copies of the solid's own cells and vertex figures. Among the others, the only patterns of a convex deltahedron that occur are those of the triangular bipyramid and the octahedron, each with unequal edges.

> **Proposition 2.** A bipyramid over an n-gon with all edges equal exists only for n ≤ 5: for n ≥ 6 the equilateral apex would lie in the plane of the polygon or the solid would cross itself. So the hexagonal and decagonal bipyramid patterns of the 16- and 600-cell have no equilateral form.

4. Variants join differently

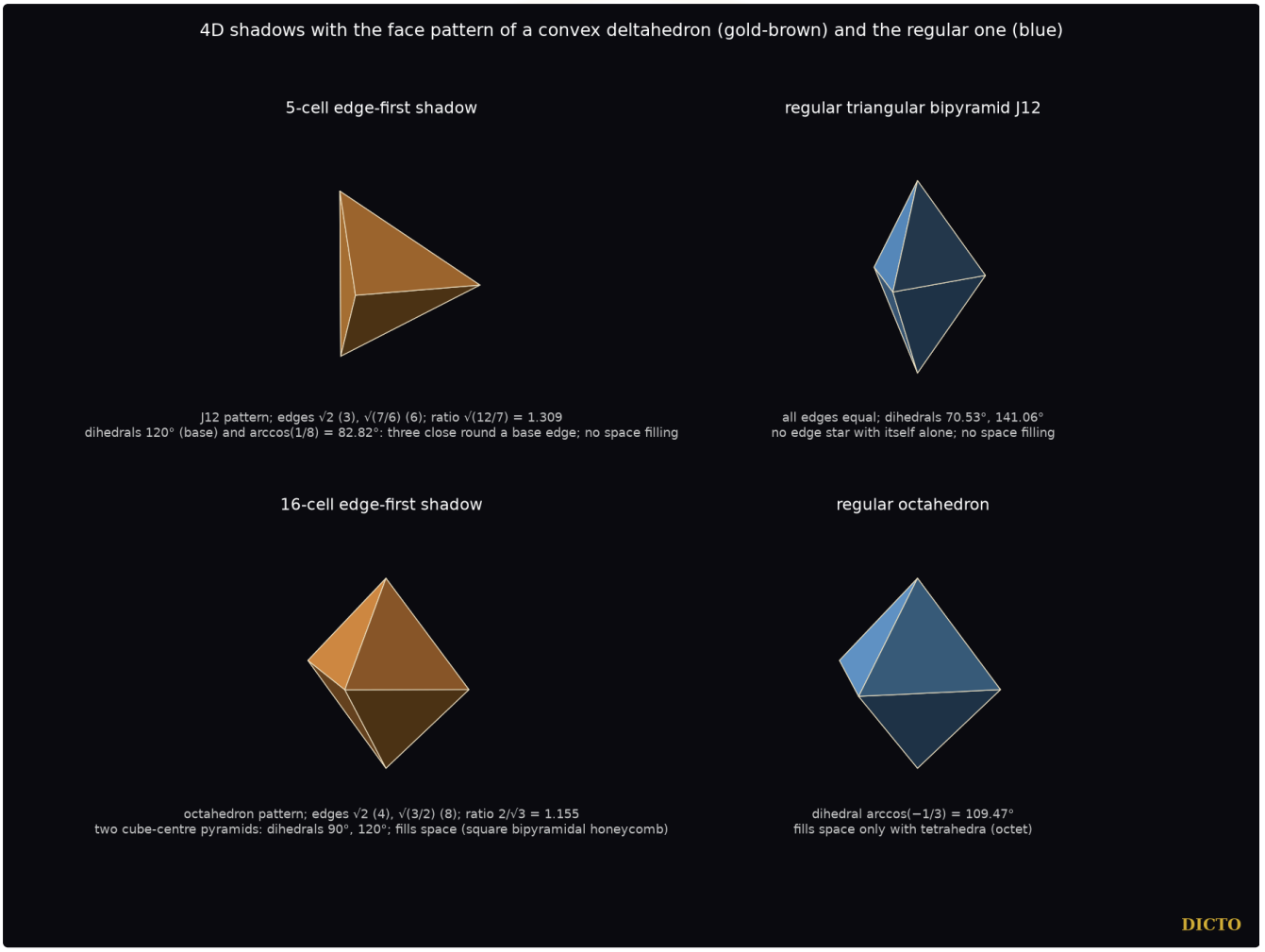


Figure 1. The two variants (gold-brown) beside their regular counterparts (blue). Top: the 5-cell's edge-first shadow, with the pattern of the triangular bipyramid J12. Bottom: the 16-cell's edge-first shadow, with the pattern of the octahedron.

TBP-5, the 5-cell's edge-first shadow, has three edges of length √2 and six of length √(7/6). Its faces match no face of J12. Its base edges have a dihedral angle of exactly 120°, so three copies of TBP-5 close face to face round a base edge; regular copies of J12 close no such ring. Its side edges have dihedral angle arccos(1/8), which closes no star of copies, so neither TBP-5 nor J12 fills space.

OCT-16, the 16-cell's edge-first shadow, is a square bipyramid: two copies of the pyramid that joins the centre of a cube to one of its faces, glued base to base. Its dihedral angles are 90° and 120°, and it fills space on its own, face to face: it is the cell of the square bipyramidal honeycomb, a known tiling (§8). The regular octahedron fills space only together with tetrahedra. The angles alone would allow a mixed star round an edge (2·90° + 70.53° + 109.47° = 360°), but the faces of OCT-16 do not match those of the regular octahedron and tetrahedron, so the variant and the regular family do not join face to face.

Both variants therefore differ from their regular counterparts in shape and in how they join, as DICTO expected.

5. The 600-cell's dented patterns, and Shadow-17

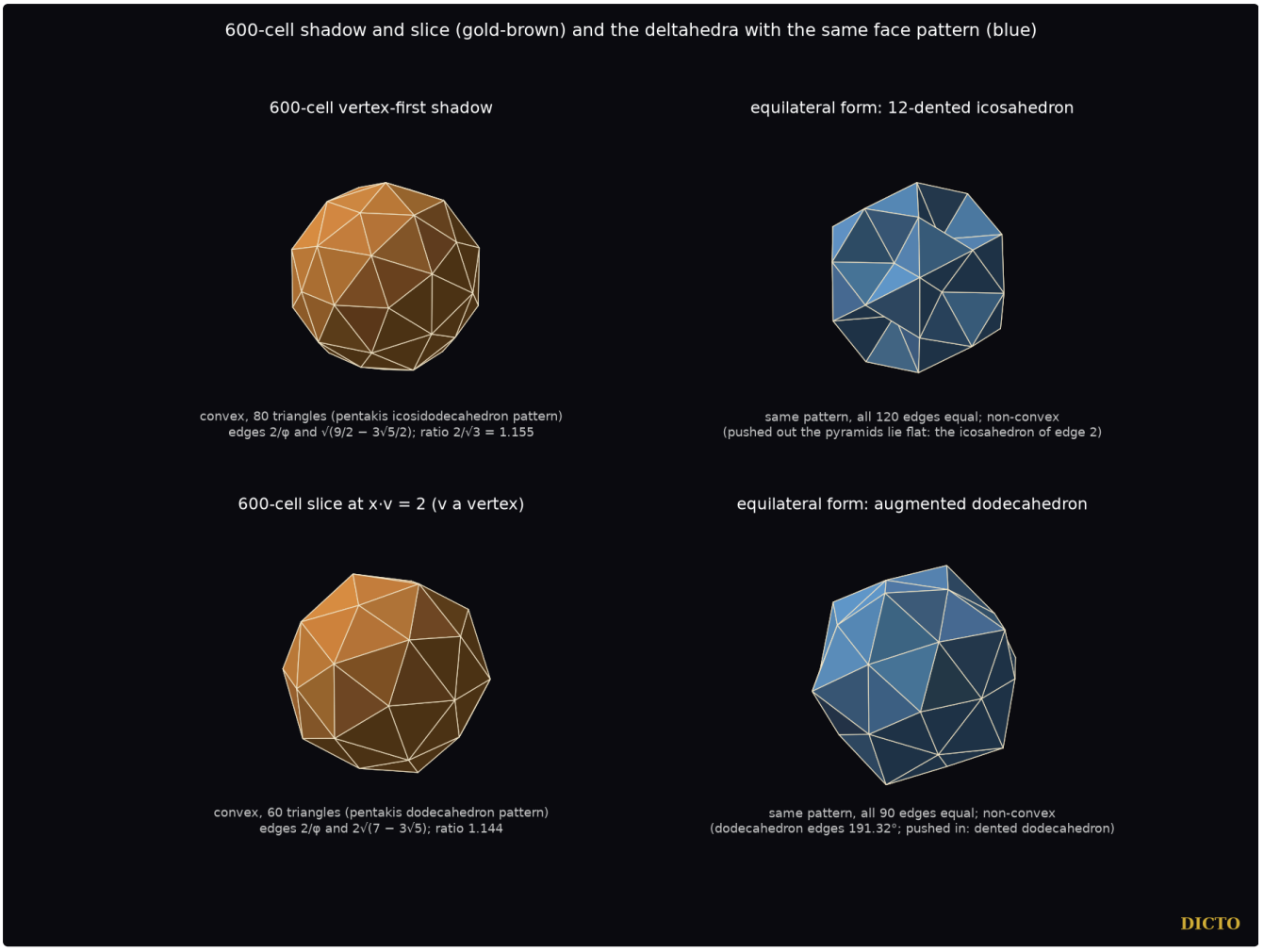


Figure 2. The 600-cell's vertex-first slices: the patterns of the pentakis icosidodecahedron and the pentakis dodecahedron, and their exact equilateral forms (12-dented icosahedron; augmented and excavated dodecahedra).

The 600-cell's vertex-first shadow is also its central slice, with 42 vertices and 80 triangles. Its equilateral form is an icosahedron with a shallow dent at each vertex (placeholder name: 12-dented icosahedron). The vertex-first slices at the dodecahedron height (32 vertices, 60 triangles) have two equilateral forms: the excavated dodecahedron, a known deltahedron, with its pyramids dented inward, and the augmented dodecahedron, with its pyramids pointing outward. The names 12-dented icosahedron and augmented dodecahedron are placeholders; both describe obvious constructions, and neither solid is claimed new.

Shadow-17. The 600-cell's edge-first slices at the outer heights have 17 vertices and 30 triangles and five-fold symmetry: a tip at each end and three rings of five vertices between them, with the vertices of the middle ring of degree 6. An equilateral form was first found numerically. DICTO's idea of cutting the pattern in halves identified it: cut at the middle ring, and each half is a J11, the gyroelongated pentagonal pyramid (the diminished icosahedron). **Shadow-17's equilateral form is two J11 glued on their pentagons.** This form is exact by construction, with every edge 1, and all 136 corner-to-corner distances of the numerical form agree with it to $4 \cdot 10^{-16}$. Its five folded edges are the edges of the shared pentagon, where the dihedral angles of the two J11 add to $2 \times 100.81° = 201.62°$. J11 is the piece with which DICTO's work began: it fills the gaps between dodecahedra and icosahedra.

6. Slice-48 and the six large slices

One of the 600-cell's six larger all-triangle slice patterns, the 48-vertex cell-first slice, has an exact equilateral form with its full symmetry: Slice-48, described at the end of this section. This section gives the method by which all six were solved and the results for each.

The six patterns have 28 to 72 vertices, with rings of high-degree vertices (7 to 10 triangles meeting at a vertex) between tips of degree 5. Cutting them in halves or caps, as for Shadow-17, matched no known piece, so each was solved directly. The exact symmetry group of each slice equals the full symmetry group of its triangle pattern. Keeping that symmetry, one vertex of each orbit is placed in the subspace fixed by its stabiliser, and each orbit of edges gives one equation $|\text{edge}|^2 = 1$; every system is square (as many equations as unknowns). The number of complex solutions of each system was found exactly over $\mathbb{Q}(\sqrt{5})$ (Singular); every solution was found and refined to 120 digits, and a system counts as complete only when the number of solutions found equals the exact count. Every real solution was then tested for a proper embedding.

Slice (600-cell)	Vertices, triangles	Symmetry	Solutions (complex / real)	Equilateral form
vertex-first, at ± $(\sqrt{5}-1)$	72, 140	Ih	8 / 4	none: vertices coincide
edge-first, at ± $(3+\sqrt{5})/2$	32, 60	D5d	128 / 48	none: edges pierce faces, or faces touch
face-first, at ± $(4+2\sqrt{5})/3$	30, 56	D3d	464 / 104	none: edges pierce faces, or faces touch
cell-first, at ± $(5+3\sqrt{5})/4$	28, 52	Td	32 / 12	none: edges pierce or cross, vertices coincide
cell-first, at ± $(3+\sqrt{5})/2$	42, 80	Td	96 / 32	none: every solution crosses itself
cell-first, at ± $(2+\sqrt{5})/2$	**48, 92**	**Td**	**192 / 48**	**Slice-48** (one shape; its two solutions differ by a half-turn)

The systems of the 72-, 32-, 28- and 42-vertex slices were also solved with only their rotation group kept, and again no form exists; the real solution counts of the same four systems were rechecked exactly.

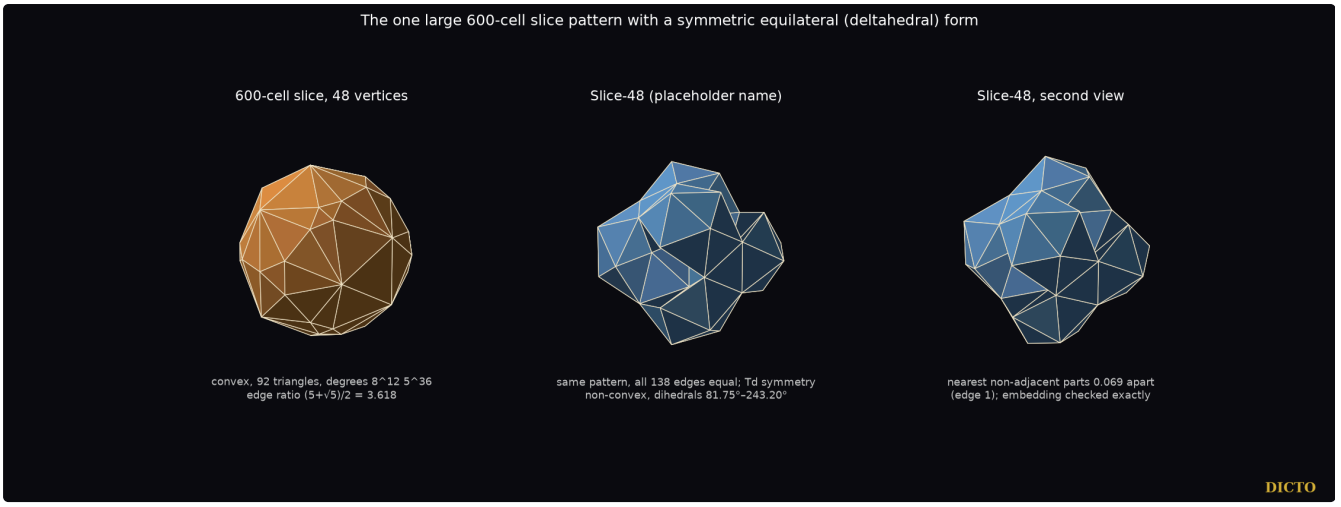


Figure 3. The 600-cell's 48-vertex cell-first slice (left, convex, edges in ratio up to (5+√5)/2) and Slice-48, its equilateral form, from two views.

> **Slice-48** (placeholder name). The 48-vertex cell-first slice of the 600-cell has exactly one equilateral form with its tetrahedral symmetry Td: 48 vertices (12 of degree 8, 36 of degree 5), 138 equal edges and 92 triangles. It is not convex: 24 of its vertices lie on its hull, and its dihedral angles are 81.75°, 130.98°, 135.83°, 143.83°, 191.03°, 197.94°, 216.17° and 243.20°. At edge length 1, its nearest non-adjacent parts are 0.069 apart; the embedding was confirmed with exact orientation tests on 85-digit rational coordinates.

Unlike Shadow-17, Slice-48 contains no piece of the icosahedron: it has no dihedral angle of 138.19° and no pentagonal cap. It is not among the non-convex deltahedra we checked (§8); since we did not read all the original papers, we do not claim it new. Forms with less symmetry than the slice's group are not excluded; the rotation-only systems of the 30- and 48-vertex slices did not finish in 50 minutes.

7. A note on the 600-cell's sections

The vertex-first "sections" of the 600-cell that are often listed — icosahedron, dodecahedron, icosahedron, icosidodecahedron — are the outlines of its vertex layers. A true cut at the dodecahedron height or at the equator also meets edges that cross it, and adds their midpoints: the true central section is the 42-vertex solid of §5, with the same pattern, edge lengths and angles as the shadow. This was checked exactly and with an independent floating-point computation.

8. Prior art and credit

Coxeter's *Regular Polytopes* treats the polytopes, their projections and the vertex-first sections of the 600-cell. The eight convex deltahedra are classical (Freudenthal and van der Waerden 1947); non-convex equilateral realisations and dented variants of deltahedral patterns are studied by Tsuruta et al. (2014). For Slice-48 we also checked the biform deltahedra (Cundy; Olshevsky), the isohedral deltahedra (Shephard; McNeill) and Trigg's spiral deltahedra, through their Wikipedia summaries. The square bipyramidal honeycomb and the excavated dodecahedron are known. In the sources checked, we did not find the comparison of the 4D shadows and slices with the deltahedra, the attachability of the variants, or the identification of Shadow-17.

Sources checked (27 October 2026; full record: study folder 2026-10-27-4d-shadows, prior-art.md, and 2026-10-27-five-pyramids, prior-art.md). *Library* (1007 references, 11 queries on deltahedra, 4D polytope projections and sections, 600-cell, Platonic face pyramids, space filling): no deltahedron literature in the library. *Web*: Tsuruta et al., ICGG 2014, deltahedral graphs (cgg.cs.tsukuba.ac.jp); Wikipedia, "600-cell", "Deltahedron", "Excavated dodecahedron"; the square bipyramidal honeycomb (via search); Polytope Wiki; MathWorld. Coxeter, *Regular Polytopes* (3rd ed. 1973) is cited for its well-known contents; it was not re-read for this paper.

9. Conclusion

Main result. The triangle-faced shadows and slices of the four-dimensional solids give deltahedra of two new kinds.
(a) **Variants that join differently.** The 5-cell's edge-first shadow (TBP-5) has the pattern of J12, and three copies of it close round an edge. The 16-cell's edge-first shadow (OCT-16) has the pattern of the octahedron, and it fills space on its own.
(b) **Equal-edged solids from the 600-cell.** Four of its shadows and slices can be rebuilt exactly with every edge the same length, keeping their pattern of triangles; all four forms are non-convex.

600-cell shadow or slice	Triangles	Rebuilt with equal edges
vertex-first shadow = central slice (42 vertices)	80	12-dented icosahedron: an icosahedron with a shallow dent at each of its 12 vertices
vertex-first slices at the dodecahedron height (32 vertices)	60	excavated dodecahedron (dents inward; known) and augmented dodecahedron (pyramids outward)
edge-first slices at the outer heights (17 vertices)	30	Shadow-17: two J11 glued on their pentagons
cell-first slices with 48 vertices	92	Slice-48: tetrahedral symmetry, dihedral angles 81.75° to 243.20°

Shadow-17 is built from two copies of J11, the piece with which DICTO's work began. Slice-48 is the largest of the four: 92 equal triangles with the full tetrahedral symmetry of its slice, found by solving all six large slices of the 600-cell exactly.

Data and code. Studies 2026-10-27-4d-shadows (q5.py exact arithmetic, polytopes.py, shadows.py, classify.py, attach.py, shadow17_j11.py, count_exact.py, solve_sage.py; data/shadows.json, all-triangle-matches.json, attachability.json, large-slices-systems.json, form_V48_full_36.json) and 2026-10-27-five-pyramids. References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; DICTO, Every Mix of the Platonic Solids Fills Space (#47), DICTO DOI 10.5281/zenodo.23551326; DICTO, Aristotle's Pyramid (#44), DICTO DOI 10.5281/zenodo.23529497.